Change of Variables in Probability

Math Foundations – Concept 28

First, an example

Before the general theory, here is the prototype. Let $X \sim \text{Uniform}(0,1)$ and define $Y = -\log X$. The transformation $g(x) = -\log x$ has inverse $g^{-1}(y) = e^{-y}$ and derivative magnitude $|g'(x)| = 1/x$. Plugging into the density-transformation rule (Theorem 1 below) with $f_X \equiv 1$ on $(0,1)$,

$$f_Y(y) \;=\; f_X(e^{-y}) \cdot e^{-y} \;=\; e^{-y}, \qquad y \geq 0.$$

Read aloud: "f sub Y of y equals f sub X of e to the minus y, times e to the minus y, which equals e to the minus y, for y greater than or equal to zero." So Y is exponential with rate 1. The factor e^{-y} is the Jacobian of the inverse map; everything that follows is a careful generalization of this single calculation.

1 Setup

Let X be a continuous random variable taking values in an open set $U \subseteq \mathbb{R}^n$ with probability density function (PDF) $f_X : U \to [0, \infty)$. Let $g : U \to V$ be a transformation; we wish to find the density of $Y = g(X)$.

Definition 1 (Diffeomorphism)**.** *A map $g : U \to V$ between open subsets of $\mathbb{R}^n$ is a C^1* diffeomorphism *if g is a bijection, $g \in C^1(U)$, and $g^{-1} \in C^1(V)$. Equivalently, g is C^1 and its Jacobian matrix $J_g(x) = (\partial g_i / \partial x_j)_{ij}$ is invertible for every $x \in U$.*

Definition 2 (Push-forward density)**.** *If X has density f_X and $Y = g(X)$ has a density, we call that density the* push-forward *of f_X along g and denote it $f_Y = g_{\#} f_X$.*

2 Main results

Theorem 1 (1D change of variables)**.** *Let X be a continuous random variable with density f_X on an open interval $(a,b) \subseteq \mathbb{R}$. Let $g : (a,b) \to (c,d)$ be a C^1 function with $g'(x) \neq 0$ for all $x \in (a,b)$. Then g is strictly monotone, hence bijective, and $Y = g(X)$ has density*

$$f_Y(y) \;=\; \frac{f_X\big(g^{-1}(y)\big)}{\big|g'\big(g^{-1}(y)\big)\big|}, \qquad y \in (c,d).$$

Read aloud: *"f sub Y of y equals f sub X of g-inverse of y, divided by the absolute value of g-prime of g-inverse of y, for y in the open interval c to d."*

Proof. Because g' is continuous and never zero on the connected interval (a,b), either $g' > 0$ everywhere or $g' < 0$ everywhere. We treat the two monotone cases via the cumulative distribution function (CDF) approach.

Case 1: g **strictly increasing.** Then $g : (a,b) \to (c,d)$ is a bijection with C^1 inverse $g^{-1} : (c,d) \to (a,b)$, and for any $y \in (c,d)$,

$$F_Y(y) \;=\; P(Y \le y) \;=\; P\big(g(X) \le y\big) \;=\; P\big(X \le g^{-1}(y)\big) \;=\; F_X\big(g^{-1}(y)\big).$$

Read aloud: "capital F sub Y of y equals the probability that Y is at most y, equals the probability that g of X is at most y, equals the probability that X is at most g-inverse of y, equals capital F sub X of g-inverse of y." Differentiating in y (chain rule; F_X is differentiable with $F_X' = f_X$, and g^{-1} is differentiable with $(g^{-1})'(y) = 1/g'(g^{-1}(y))$ by the inverse function theorem),

$$f_Y(y) \;=\; F_Y'(y) \;=\; f_X\big(g^{-1}(y)\big)\,\big(g^{-1}\big)'(y) \;=\; \frac{f_X\big(g^{-1}(y)\big)}{g'\big(g^{-1}(y)\big)}.$$

Read aloud: "f sub Y of y equals capital F sub Y prime of y, equals f sub X of g-inverse of y, times the derivative of g-inverse at y, equals f sub X of g-inverse of y divided by g-prime of g-inverse of y." Since $g' > 0$, we have $|g'| = g'$, so this equals the claimed formula.

Case 2: g **strictly decreasing.** Then for $y \in (c,d)$,

$$F_Y(y) \;=\; P\big(g(X) \le y\big) \;=\; P\big(X \ge g^{-1}(y)\big) \;=\; 1 - F_X\big(g^{-1}(y)\big).$$

Read aloud: "capital F sub Y of y equals the probability that g of X is at most y, equals the probability that X is at least g-inverse of y, equals one minus capital F sub X of g-inverse of y." Differentiating,

$$f_Y(y) \;=\; -\,f_X\big(g^{-1}(y)\big)\,\big(g^{-1}\big)'(y) \;=\; -\,\frac{f_X\big(g^{-1}(y)\big)}{g'\big(g^{-1}(y)\big)}.$$

Read aloud: "f sub Y of y equals negative f sub X of g-inverse of y times the derivative of g-inverse at y, equals negative f sub X of g-inverse of y divided by g-prime of g-inverse of y." Since $g' < 0$, $-1/g' = 1/|g'|$, recovering the claimed formula.

In both cases, $f_Y(y) = f_X(g^{-1}(y))/|g'(g^{-1}(y))|$, completing the proof. □

Theorem 2 (Multivariate change of variables)**.** *Let X be a continuous random vector on an open set $U \subseteq \mathbb{R}^n$ with density f_X, and let $g : U \to V$ be a C^1 diffeomorphism onto an open set $V \subseteq \mathbb{R}^n$. Then $Y = g(X)$ has density*

$$f_Y(y) \;=\; f_X\big(g^{-1}(y)\big)\cdot\big|\det J_{g^{-1}}(y)\big| \;=\; \left.\frac{f_X(x)}{\big|\det J_g(x)\big|}\right|_{x=g^{-1}(y)}, \qquad y \in V.$$

Read aloud: *"f sub Y of y equals f sub X of g-inverse of y, times the absolute value of the determinant of the Jacobian of g-inverse at y, which also equals f sub X of x divided by the absolute value of the determinant of the Jacobian of g at x, evaluated at x equals g-inverse of y, for y in V."*

Proof sketch. For any Borel set $B \subseteq V$, using $g^{-1}(B) \subseteq U$ and the multivariate substitution theorem from analysis,

$$P(Y \in B) = P(X \in g^{-1}(B)) = \int_{g^{-1}(B)} f_X(x)\,dx = \int_B f_X(g^{-1}(y))\,|\det J_{g^{-1}}(y)|\,dy.$$

Read aloud: "the probability that Y lies in B equals the probability that X lies in g-inverse of B, equals the integral over g-inverse of B of f sub X of x, d-x, equals the integral over B of f sub X of g-inverse of y, times the absolute value of the determinant of the Jacobian of g-inverse at y, d-y." Because B was arbitrary, the integrand is the density of Y. □

3 Canonical example

Let $X \sim \text{Uniform}[0,1]$ and $Y = -\log X$. Then $g(x) = -\log x$ is C^1 on $(0,1)$ with $g'(x) = -1/x$, $|g'(x)| = 1/x$, and $g^{-1}(y) = e^{-y}$. By Theorem 1,

$$f_Y(y) \;=\; \frac{1}{1/e^{-y}} \;=\; e^{-y}, \qquad y > 0,$$

Read aloud: "f sub Y of y equals one divided by one over e to the minus y, which equals e to the minus y, for y greater than zero." i.e. $Y \sim \text{Exp}(1)$. This identity underlies inverse-CDF sampling.